

Artificial Intelligence in Hematology: Can Algorithms Replace the First Glance at the Blood Smear?



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Introduction

Peripheral blood smear examination remains a fundamental component of hematological diagnosis. It provides direct morphological information on **red blood cells, leukocytes and platelets**, and may reveal abnormalities that cannot be fully characterized by automated CBC alone. However, conventional microscopy is: time-consuming; dependent on operator expertise; subject to inter-observer variability; difficult to standardize; particularly challenging for rare and morphologically complex cells. Recent advances in **digital microscopy, computer vision, machine learning and deep learning** have enabled automated localization, segmentation and classification of blood cells. Several platforms are now available for routine laboratory use, including Sysmex DI-60, CellaVision systems and Mindray MC-80.

The major question is therefore no longer whether AI can recognize blood cells, but whether it can reliably replace the initial human assessment of a peripheral blood ?



objectives

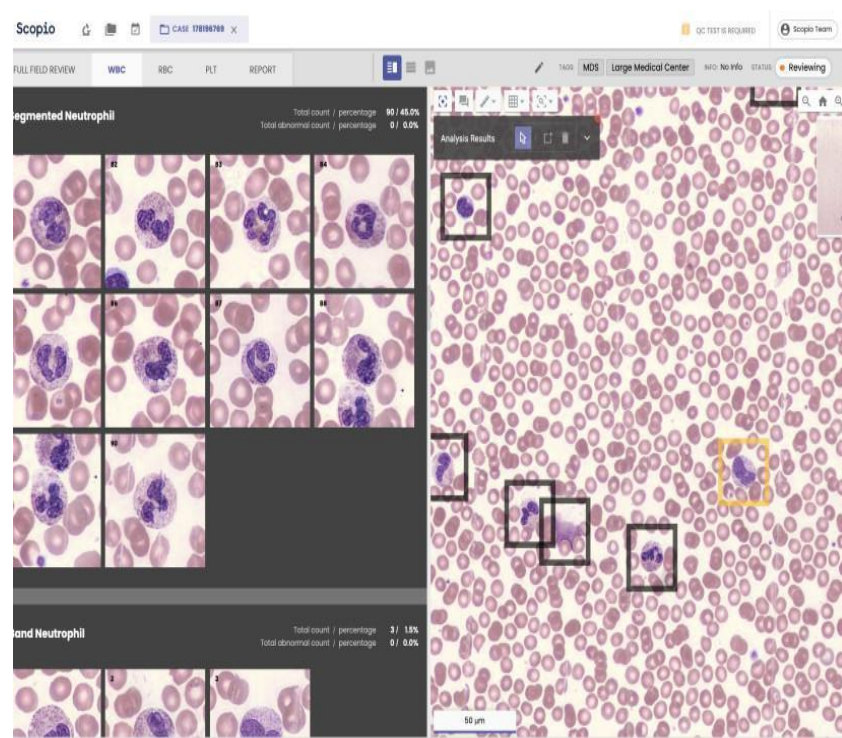
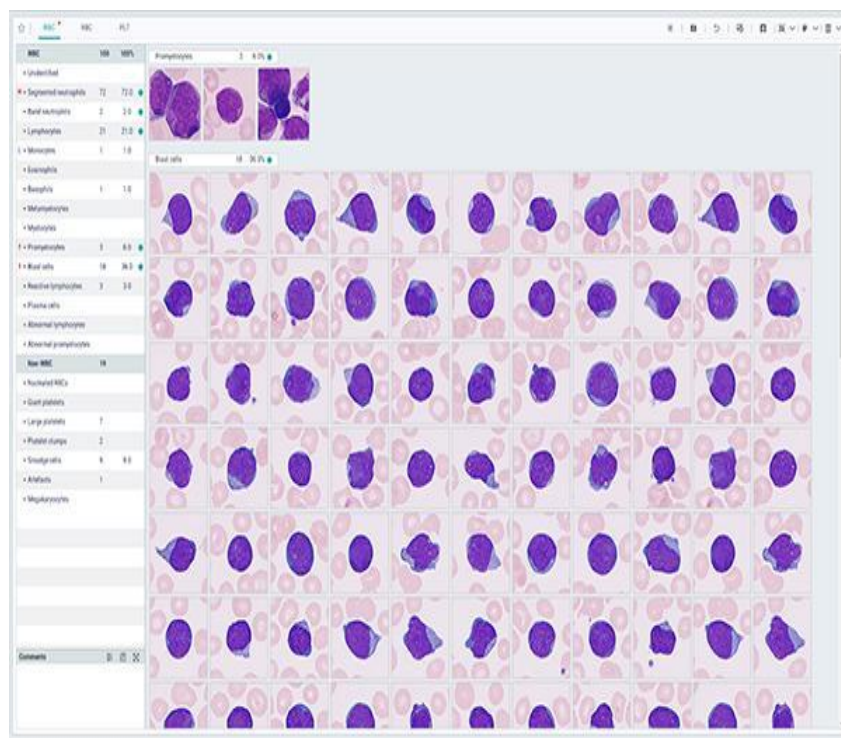
Review the current applications of **artificial intelligence and automated digital morphology** in peripheral blood smear analysis, focusing on:

- Automated cell localization and classification
- Detection of abnormal cells
- Recognition of blasts and atypical lymphocytes
- Red-cell morphological analysis
- Diagnostic performance
- Reduction of manual workload
- Current limitations and the role of human expertise



Methods and Materials

A narrative literature review was conducted using recent peer-reviewed studies addressing artificial intelligence, machine learning, deep learning and digital morphology in hematology. Particular attention was given to automated cell classification, detection of abnormal cells and AI-assisted diagnosis. The main comparator was conventional manual microscopy, performed by experienced laboratory professionals. The reviewed studies evaluated: Correlation → Sensitivity → Specificity → Accuracy → Agreement → Reproducibility → Workflow



results and discussion

Current AI-assisted platforms, including **Sysmex DI-60, CellaVision and Mindray MC-80**, provide automated cell localization, pre-classification and morphological characterization. .

Mindray MC-80 — AI-assisted morphology:

The Mindray MC-80 represents one of the newer AI-based digital morphology platforms. A large 2026 validation study included 1,188 routine blood samples, of which 881 were normal and 307 abnormal. MC-80 preclassification was compared with hematology analyzer results for normal samples and expert manual microscopy for abnormal samples.

Main findings:

- Correlation coefficients >0.90 for the majority of leukocyte subsets
- Overall specificity >99%
- Sensitivity >90% for most abnormal cell categories
- Plasma-cell sensitivity: 75% when ≥1% plasma cells were present
- Lower performance was observed for some abnormal populations, Particularly blasts and reactive lymphocytes.

Interpretation: The MC-80 demonstrates excellent potential for routine classification, but expert verification remains necessary for clinically important abnormal cells.

Sysmex DI-60 :

The Sysmex DI-60 is an automated digital cell-locating and image-analysis system. In a study of 822 peripheral blood smears, the DI-60 was compared with manual slide review for WBC differentials and RBC morphology. Overall agreement between preclassification and user-verified WBC classification was 89.4%.

RBC morphology : The system demonstrated acceptable performance for: polychromasia; target cells; ovalocytes. However, schistocyte recognition illustrated an important limitation: (Sensitivity: 97.2%Specificity: 10.4%) This means that a system can be highly sensitive while generating many false-positive classifications for a morphologically difficult category.

Abnormal WBC detection :Another analytical validation reported: Sensitivity for immature/pathological cells: 88.7% ; Specificity: 83.0% and AUC: 0.86. The authors concluded that light microscopy remained mandatory in doubtful cases.

Cella Vision:

CellaVision systems digitize blood-smear images and perform automated preclassification, followed by human verification or reclassification. In a study of 359 peripheral blood smears using CellaVision DM96, correlation with manual microscopy varied according to cell type:

Highest correlation: $r^2 = 0.99$ for blasts.

Lowest reported correlation: $r^2 = 0.72$ for metamyelocytes.

The study emphasized that performance depended strongly on the cell type, with lower correlation generally occurring for rarer cells.

CellaVision DC-1: In a comparison involving 100 normal and 100 abnormal blood smears, the DC-1 demonstrated: Sensitivity: 96.6%; Specificity: 95.3% when compared with manual microscopy.

However, another evaluation showed that positive predictive performance was particularly problematic for several rare categories, including: metamyelocytes; myelocytes; promyelocytes; blasts; plasma cells; nucleated RBCs.

System	Technology / application	Main use
Sysmex DI-60	Automated digital morphology	WBC differential + RBC morphology
CellaVision	Digital microscopy + automated preclassification	WBC/RBC morphology
Mindray MC-80	AI-based digital morphology	Normal and abnormal cell classification
Other emerging AI models	Deep learning / computer vision	Automated cell recognition

COMPARISON WITH CONVENTIONAL MICROSCOPY		
Parameter	Manual microscopy	AI-assisted digital morphology
Speed	Relatively slow	High potential for workflow optimization
Routine cell classification	Excellent	Excellent
Standardization	Operator-dependent	High
Reproducibility	Variable	Generally high
Digital image archive	No	Yes
Remote review	Limited	Possible
Rare cells	Strong with expert	Still challenging
Atypical cells	Expert-dependent	Variable
Clinical interpretation	Excellent	Limited
Final validation	Human	Human remains essential

Conclusion

Artificial intelligence is transforming peripheral blood smear examination from conventional subjective microscopy toward standardized digital morphology. Current systems such as Mindray MC-80, Sysmex DI-60 and CellaVision demonstrate strong performance for routine cell classification and promising capabilities for abnormal-cell detection. However, rare, atypical and morphologically complex cells remain challenging. Therefore, current evidence supports an AI-assisted rather than AI-replaced model. AI should be considered a powerful second reader that improves efficiency, standardization and workflow, while the laboratory specialist remains essential for morphological verification, clinical correlation and final interpretation.

Bibliographies

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